Deep Learning CIA (2087101)

Undercomplete and Denoising Autoencoders on Fashion-MNIST

Student name: _____ **Roll number:** _____

Framework: PyTorch **Dataset:** Fashion-MNIST, 28 × 28 grayscale images

1. Objective

Design an undercomplete autoencoder to compress and reconstruct Fashion-MNIST images, evaluate test reconstruction Mean Squared Error (MSE), compare bottleneck sizes 8, 32, and 64, and train a denoising autoencoder that maps noisy images to clean targets.

2. Architecture

```
Input image (28×28 = 784 values)
        |
Encoder: 784 → 256 → 128 → latent z
        |
Bottleneck: z ∈ R^k, k ∈ {8, 32, 64}
        |
Decoder: k → 128 → 256 → 784
        |
Reconstructed image (28×28)
```

ReLU activations are used in hidden layers and sigmoid in the output layer. The model is trained with MSE loss and Adam. For the denoising experiment, Gaussian noise is added to the input, while the target remains the clean image.

3. Metrics and results

Latent size	Dimensional compression ratio	Test MSE (fill after run)
8	784 / 8 = 98:1	_____
32	784 / 32 = 24.5:1	_____
64	784 / 64 = 12.25:1	_____
Denoising, k=32	24.5:1 (latent dimensions)	_____

Compression ratio here means input dimensionality divided by latent dimensionality. It is not a true on-disk file-size ratio because model weights and encoding overhead are excluded. Insert the measured test MSE values from the executed notebook before submission.

4. Analysis

A smaller bottleneck provides stronger dimensional compression but has less capacity to preserve image detail, so it often produces higher reconstruction error. Larger bottlenecks can preserve more information but reduce the compression ratio. These are expected trends; the actual MSE values should be taken from the notebook run. Denoising performance should be assessed by comparing reconstructed images with clean targets and reporting the noisy-input test MSE.

Regularization extension: the notebook supports weight decay (L2 regularization) through the `weight_decay` parameter. To discuss its effect, train a comparable model with a nonzero weight decay such as 1e-4 and compare test MSE.

5. Figures to include

The notebook generates (a) training loss curves, (b) loss curves for latent sizes 8, 32, and 64, and (c) a three-row visual grid of original, noisy, and reconstructed images. These figures should be visible in the saved, executed notebook.

6. Conclusion

The experiments demonstrate how an undercomplete autoencoder learns a compact latent representation and how bottleneck size affects the trade-off between compression and reconstruction fidelity. The denoising autoencoder learns to reconstruct clean targets from corrupted inputs. Final conclusions should cite the actual test MSE values and visual outputs obtained by running the notebook.

Submission reminder: execute all notebook cells, fill the results table above with observed values, and upload the notebook, README, output logs, and this report to the GitHub repository.